MTH213 | MAT213 – Differential Equations I

Review Questions on Second-order Differential Equations and Systems of First-order differential Equations

Question 1

For each of the following differential equations, state which method is more appropriate — Undetermined Coefficients (UC) or Variation of Parameters (VoP) — and justify your choice briefly.
(Do not solve the equations.)

I. $y'' + 9y = 4\sin\ 3x$

II. $y'' - 2y = e^x \ln\ x$

III. $y'' + y = 3x - x^2$

IV. $y'' + 4y = \sec\ x$

Question 2

Solve the following boundary value problem using the method of undetermined coefficients:

$$y'' + 4y = \cos\ 2x, y(0) = 0, y\left(\frac{\pi}{4}\right) = \frac{1}{2}.$$

Question 3

I. Define the Wronskian of two functions $y_1(x)$ and $y_2(x)$.

II. Show that the Wronskian of e^x and e^{-x} is constant and non-zero.
Hence, explain what this implies about the linear independence of the two functions.

Question 4

A mass of 1.5 kg is attached to a horizontal spring with spring constant 24 N/m.
The system is free of damping and external forces.

At time $t = 0$, the mass is displaced 0.2 m from its equilibrium position and given an initial velocity of 0.1 m/s toward equilibrium.

Find the displacement $x(t)$ of the mass from equilibrium at any time t.

Question 5

I. Write the following fourth-order differential equation as a system of first-order linear differential equations:

$$y^{(4)} + 2y'' + 5y = e^t$$

subject to the initial conditions

$$y(0) = 0, y'(0) = 1, y''(0) = 0, y'''(0) = 2.$$

II. Convert the following second-order differential equation into an equivalent system of first-order differential equations.
Solve the resulting system and hence obtain the general solution of the original differential equation:

$$y'' - 4y' + 4y = 0.$$